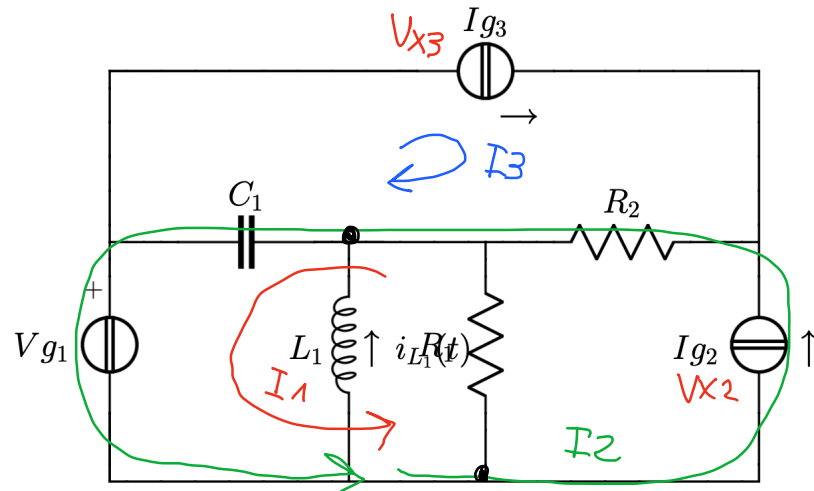


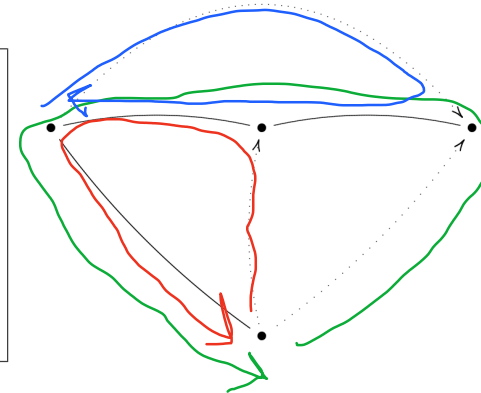
Parallelo Maglie

sabato 2 agosto 2025 13:21

Risolvere il circuito in figura



$$\begin{aligned} \mathbf{V}_{g1} &= 2 - 2j \\ R_1 &= 1 \\ L_1 &= \frac{1}{2} \\ \mathbf{I}_{g2} &= -j \\ C_1 &= \frac{1}{2} \\ R_2 &= 1 \\ \mathbf{I}_{g3} &= -1 - j \\ \omega &= 2 \end{aligned}$$



$$Y_a = \frac{1}{j\omega L_1} + \frac{1}{R_1} = \frac{1}{j} + 1 = 1 - j$$

$$Z_a = \frac{1}{Y_a} = \frac{1}{1 - j} = \frac{1 + j}{2} = \frac{1}{2} + \frac{j}{2}$$

Ormai ometto le descrizioni ai passaggi, a questo punto dovrebbe essere chiaro.

$$\begin{bmatrix} \frac{1}{j\omega C_1} + Z_a & \frac{1}{j\omega C_1} & \frac{1}{j\omega C_1} \\ \frac{1}{j\omega C_1} & \frac{1}{j\omega C_1} + R_2 & \frac{1}{j\omega C_1} + R_2 \\ \frac{1}{j\omega C_1} & \frac{1}{j\omega C_1} + R_2 & \frac{1}{j\omega C_1} + R_2 \end{bmatrix} \begin{pmatrix} I_1 \\ I_2 \\ I_3 \end{pmatrix} = \begin{pmatrix} -V_{g1} \\ V_{x2} - V_{g1} \\ V_{x3} \end{pmatrix}$$

$$\begin{pmatrix} 1 & -1 & 0 \\ -1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} I_1 \\ I_2 \\ I_3 \end{pmatrix} = \begin{pmatrix} -2 + 2j \\ 0 \\ -1 - j \end{pmatrix}$$

$$\begin{cases} \left(\frac{1}{j\omega C_1} + 2\Omega \right) I_1 + \frac{1}{j\omega C_1} I_2 + \frac{1}{j\omega C_1} I_3 = -V_{g1} \\ \frac{1}{j\omega C_1} I_1 + \left(\frac{1}{j\omega C_1} + R_2 \right) I_2 + \left(\frac{1}{j\omega C_1} + R_2 \right) I_3 = V_{x2} - V_{g1} \\ \frac{1}{j\omega C_1} I_1 + \left(\frac{1}{j\omega C_1} + R_2 \right) I_2 + \left(\frac{1}{j\omega C_1} + R_2 \right) I_3 = V_{x3} \\ I_2 = I_{g2} \\ I_3 = I_{g3} \end{cases}$$

Risolviamo il sistema

$$\begin{cases} \left(-j + \frac{1}{2}j + \frac{1}{2} \right) I_1 - j I_2 - j I_3 = 2j - 2 \\ -j I_1 + (1-j) I_2 + (1-j) I_3 = V_{x2} + 2j - 2 \\ -j I_1 + (1-j) I_2 + (1-j) I_3 = V_{x3} \\ I_2 = -j \\ I_3 = -1-j \end{cases}$$

$$\begin{cases} \left(-\frac{1}{2}j + \frac{1}{2} \right) I_1 - 1 + j - 1 = 2j - 2 \\ -j I_1 - j - 1 - 2 = V_{x2} + 2j - 2 \\ -j I_1 - j - 1 - 2 = V_{x3} \end{cases} \rightarrow \begin{cases} \left(\frac{1-j}{2} \right) I_1 = j \quad (1) \\ -j I_1 - j - 1 - 2 = V_{x2} + 2j - 2 \quad (2) \\ -j I_1 - j - 1 - 2 = V_{x3} \quad (3) \end{cases}$$

Dalla (1) Ricavo I_1

$$I_1 = \frac{2j}{1-j} \cdot \frac{1+j}{1+j} = \frac{2j-2}{2} = j-1$$

valore nella (2)

$$\cancel{1+j} - \cancel{j} - \cancel{1} - \cancel{2} + \cancel{2} - 2j = V_{x2}$$

Soluzioni sistema

$$\begin{cases} I_1 = j-1 \quad \checkmark \\ I_2 = -j \quad \checkmark \end{cases}$$

$$V_{X2} = -2.5$$

$$1 + 5 - 5 - 1 - 2 = V_{X3}$$

$$V_{X3} = -2$$

$$\begin{cases} I_3 = -1 - 5 \\ V_{X2} = -2.5 \\ V_{X3} = -2 \end{cases}$$

BILANCIO DI POTENZA

$$I_{C1} = -I_2 - I_3 - I_1 = 5 + 1 + 5 - 5 + 1 = 5 + 2$$

$$I_{L1} + I_{R1} = I_1 = 5 - 1$$

$$I_{R2} = -I_2 - I_3 = 2.5 + 1$$

$$I_{g3} = -1 - 5 \text{ (NOTO)}$$

$$I_{g2} = -5 \text{ (NOTO)}$$

$$I_{L1} = \frac{V_{L1}}{3\omega L_1} = \frac{-1}{5} = 5$$

$$I_{R1} = \frac{V_{R1}}{R_1} = \frac{-1}{1} = -1$$

$$I_{g1} = -I_2 - I_1 = 5 + 1 - 5 = 1$$

$$V_{C1} = \frac{1}{5\omega C_1} I_{C1} = \frac{1}{5} (5 + 2) = 1 - 2.5$$

$$V_{L1} = V_{R1} = Z_{\omega} \cdot (I_{L1} + I_{R1}) = \left(\frac{1}{2} + \frac{1}{2}j\right) (5 - 1) = -1$$

$$V_{R2} = R_2 \cdot I_{R2} = 1 \cdot 2.5 + 1 = 2.5 + 1$$

$$V_{g3} = V_{X3} = -2$$

$$V_{g2} = V_{X2} = -2.5$$

$$V_{g1} = 2 - 2.5 \text{ (NOTO)}$$

$$P_{C|g2} = \frac{1}{2} V_{g2} I_{g2}^* = \frac{1}{2} (-2.5) 5 = +1.25$$

$$P_{C|g3} = \frac{1}{2} V_{g3} I_{g3}^* = \frac{1}{2} (-2) (-1 + 5) = 1 - 3$$

$$P_{C|g1} = \frac{1}{2} V_{g1} I_{g1}^* = \frac{1}{2} (2 - 2.5) 1 = 1 - 3$$

$$P_{A|R1} = \frac{1}{2} R_1 |I_{R1}|^2 = \frac{1}{2} \cdot 1 \cdot 1 = \frac{1}{2}$$

$$\rightarrow 1 + 1 - 1 = \frac{1}{2} + \frac{5}{2}$$

$$3 = 3$$



$$P_{R2} = \frac{1}{2} R_2 |I_{R2}|^2 = \frac{1}{2} \cdot 1 \cdot 5 = \frac{5}{2}$$

$$Q_{L1} = \frac{1}{2} \omega L_1 |I_{L1}|^2 = \frac{1}{2} \cdot 2 \cdot \frac{1}{2} (1) = \frac{1}{2}$$

$$Q_{C1} = \frac{1}{2} \omega C_1 |V_{C1}|^2 = -\frac{1}{2} \cdot 2 \cdot \frac{1}{2} (5) = -\frac{5}{2}$$

$$P_{Rg1} = \text{Im}(P_{C1g1}) = -1$$

$$P_{Rg2} = \text{Im}(P_{C1g2}) = 0$$

$$P_{Rg3} = \text{Im}(P_{C1g3}) = -1$$

$$\rightarrow -\frac{5}{2} + \frac{1}{2} = -1 - 1$$

$$-2 = -2$$

